Accelerated Observers and Born Rigidity

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Scope

These notes conclude the discussion of accelerated observers in flat spacetime, in preparation for the study of radiation from moving charges.

The goals are:

- understand why Rindler coordinates are special;
- distinguish accelerated congruences from Born-rigid congruences;
- construct accelerated coordinates with identical proper acceleration.

Rindler Coordinates Revisited

Consider Minkowski spacetime with coordinates (T, X, Y, Z) and metric

$$ds^2 = -dT^2 + dX^2 + dY^2 + dZ^2.$$

Here $T \equiv x^0 = ct$ for ordinary time t, i.e. time measured in length units, consistent with the conventions fixed for this course; a proper acceleration α quoted below in these units corresponds to $\alpha_{\mathrm{SI}} = c^2\alpha$ in SI units.

Define

$$T = \rho\sinh(a\eta), \qquad X = \rho\cosh(a\eta).$$

Substituting,

$$ds^2 = -(a\rho)^2 d\eta^2 + d\rho^2 + dY^2 + dZ^2.$$

Worldlines $\rho = \mathrm{const}$ satisfy

$$X^2 - T^2 = \rho^2.$$

These are hyperbolae.

Using proper time,

$$d\tau = a\rho\, d\eta,$$

the four-velocity is

$$u^\mu = \left(\cosh\frac{\tau}{\rho}, \sinh\frac{\tau}{\rho}\right),$$

and the four-acceleration is

$$a^\mu = \frac{1}{\rho}\left(\sinh\frac{\tau}{\rho}, \cosh\frac{\tau}{\rho}\right).$$

Hence

$$a^\mu a_\mu = \frac{1}{\rho^2}.$$

The proper acceleration is

$$\alpha = \frac{1}{\rho}.$$

> **Note**
>
> Different Rindler observers have different proper accelerations. The observer closest to the horizon accelerates the most.

Born Rigidity

For a congruence with four-velocity u^μ, define

$$h_{\mu\nu} = \eta_{\mu\nu} + u_\mu u_\nu.$$

The spatial deformation tensor is

$$\Theta_{\mu\nu} = h_\mu{}^\alpha h_\nu{}^\beta \nabla_{(\alpha} u_{\beta)}.$$

Born rigidity requires

$$\Theta_{\mu\nu} = 0.$$

This means that neighboring observers maintain constant proper separation in their instantaneous rest frame.

Rindler observers satisfy this condition.

Why Equal Acceleration is Incompatible with Born Rigidity

Suppose every observer had the same proper acceleration,

$$a^\mu a_\mu = a_0^2.$$

Imagine two neighboring observers separated by a fixed proper distance.

After a short interval, the rear observer must have acquired a slightly larger velocity than the front observer if the separation is to remain fixed in the instantaneous rest frame.

Therefore the proper acceleration must vary along the congruence.

The result is that translational Born-rigid acceleration necessarily requires an acceleration gradient.

Rindler coordinates realize exactly this behavior.

Accelerated Coordinates with Equal Proper Acceleration

Start from the uniformly accelerated trajectory

$$T_0(\tau) = \frac{1}{a}\sinh(a\tau), \qquad X_0(\tau) = \frac{1}{a}\cosh(a\tau).$$

Translate the trajectory by a constant spatial label ξ:

$$T(\tau,\xi) = \frac{1}{a}\sinh(a\tau),$$

$$X(\tau,\xi) = \xi + \frac{1}{a}\cosh(a\tau).$$

Every worldline has

$$a^\mu a_\mu = a^2.$$

The different observers therefore possess identical proper acceleration.

Computing

$$dT = \cosh(a\tau)d\tau,$$

$$dX = d\xi + \sinh(a\tau)d\tau,$$

gives

$$ds^2 = -d\tau^2 + 2\sinh(a\tau)d\tau d\xi + d\xi^2.$$

The metric has

$$g_{0i} \neq 0.$$

The congruence is accelerated but not of the Rindler type.

Comparison with Rindler

Property	Rindler	Equal-Acceleration Family
Proper acceleration	varies as $1/\rho$	constant
Born rigid	yes	no
Horizon	yes	no
g_{0i}	0	nonzero
Stationary	yes	no

Transition to Electrodynamics

Accelerated observers reveal nontrivial geometric effects. Accelerated charges reveal something more dramatic: they radiate. The energy-momentum tensor needed to make that statement precise, and the derivation of the radiation field itself, are developed in Energy of the Electromagnetic Field and Radiation from a Moving Charge.

Summary

The Rindler congruence is a special Born-rigid accelerated frame whose proper acceleration varies spatially.

Uniform proper acceleration for all observers can be achieved, but only at the cost of losing Born rigidity.

These accelerated worldlines are exactly the ones whose radiation we study next.

Exercises

Exercise 1: Kinematics of the Rindler Congruence (Essential)

a) Starting from $T = \rho \sinh(a\eta)$, $X = \rho \cosh(a\eta)$ and $\mathrm{d}\tau = a\rho\, \mathrm{d}\eta$, compute $\mathrm{d}T/\mathrm{d}\tau$ and $\mathrm{d}X/\mathrm{d}\tau$ directly and confirm $u^\mu = (\cosh(\tau/\rho), \sinh(\tau/\rho))$.

b) Differentiate u^μ with respect to τ to obtain a^μ, and verify explicitly that $a^\mu a_\mu = 1/\rho^2$.

c) Using the u^μ found in (a), check that $X^2 - T^2 = \rho^2$ is preserved along the worldline, i.e. that $\mathrm{d}(X^2 - T^2)/\mathrm{d}\tau = 0$.

Exercise 2: Born Rigidity of the Rindler Congruence (Essential)

The notes assert that "Rindler observers satisfy this condition" ($\Theta_{\mu\nu} = 0$) without showing why. This exercise fills that gap.

a) In the coordinates (η, ρ, Y, Z), write down the spatial part of

$$ds^2 = -(a\rho)^2 \mathrm{d}\eta^2 + \mathrm{d}\rho^2 + \mathrm{d}Y^2 + \mathrm{d}Z^2$$

(the coefficients of $\mathrm{d}\rho^2$, $\mathrm{d}Y^2$, $\mathrm{d}Z^2$) and note that none of them depend on η, and that there are no $\mathrm{d}\eta\, \mathrm{d}\rho$, $\mathrm{d}\eta\, \mathrm{d}Y$, $\mathrm{d}\eta\, \mathrm{d}Z$ cross terms.

b) Explain why a spatial metric between neighboring $\rho = \text{const}$ worldlines that is independent of η, with no such cross terms, means those worldlines maintain constant proper separation for all η — i.e. $\Theta_{\mu\nu} = 0$.

c) Contrast this with the equal-acceleration family of the next section, where a term $2\sinh(a\tau)\, \mathrm{d}\tau\, \mathrm{d}\xi$ appears in the metric: explain why this cross term signals a departure from Born rigidity.

Exercise 3: Kinematics of the Equal-Acceleration Family

a) Starting from $T(\tau, \xi) = \frac{1}{a}\sinh(a\tau)$, $X(\tau, \xi) = \xi + \frac{1}{a}\cosh(a\tau)$, compute $u^\mu = \partial(T, X)/\partial\tau$ at fixed ξ and $a^\mu = \partial u^\mu/\partial\tau$, and verify explicitly that $a^\mu a_\mu = a^2$, independent of ξ.

b) Using $\mathrm{d}T = \cosh(a\tau)\mathrm{d}\tau$ and $\mathrm{d}X = \mathrm{d}\xi + \sinh(a\tau)\mathrm{d}\tau$ as given in the notes, substitute into $ds^2 = -\mathrm{d}T^2 + \mathrm{d}X^2 + \mathrm{d}Y^2 + \mathrm{d}Z^2$ and verify term by term that

$$ds^2 = -\mathrm{d}\tau^2 + 2\sinh(a\tau)\, \mathrm{d}\tau\, \mathrm{d}\xi + \mathrm{d}\xi^2.$$

c) Identify which metric component is responsible for the loss of Born rigidity found in Exercise 2.

Exercise 4: Comparing the Two Congruences

a) In the Rindler congruence, the proper acceleration $\alpha = 1/\rho$ genuinely differs between worldlines at different ρ, while in the equal-acceleration family every worldline has the same $a^\mu a_\mu = a^2$ regardless of ξ. Using the results of Exercises 2 and 3, explain in your own words which of these two behaviors is compatible with $\Theta_{\mu\nu} = 0$, and why.

b) The qualitative argument in "Why Equal Acceleration is Incompatible with Born Rigidity" states that the rear observer of a rigid pair "must have acquired a slightly larger velocity than the front observer." Identify which metric component found in Exercise 3(b) encodes exactly this effect.